Eitmam Omar Sanam

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

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Academic Credentials

2023 – 2027 (exp.) B.Sc. in Electrical and Electronic Engineering
Rajshahi University of Engineering & Technology (RUET)
CGPA: 3.70/4.00 (through 7th semester)

Research Interests

Generative AI for Medicine, Trustworthy and Explainable AI, Medical Report Generation, Embedded AI and TinyML, Human Activity Recognition, Biomedical Signal Processing, and Wearable Devices.

Research Experience

2025 – Present **Research Assistant,** [Lab Profile]
SPML Lab, EEE, RUET

- Built embedded AI systems on STM32 (Cortex-M7), Arduino Nano 33 BLE Sense and Raspberry Pi, with HAL firmware in STM32CubeIDE, on-hardware debugging and real-board power profiling.
- Gained hands-on experience in embedded AI, low-resource AI, and biomedical signal processing.
- Authored multiple technical research papers submitted to reputed journals.

Research Publications

Under Review **Eitmam Omar Sanam** and Mohammod Abdul Motin, "EdgeSSM-HAR: On-Device Personalized State-Space Models for Human Activity Recognition in Consumer Wearables," ***IEEE Transactions on Consumer Electronics***.

Under Review **Eitmam Omar Sanam** and Mohammod Abdul Motin, "RepSenHAR: Structural Reparameterization for Inference-Efficient Wearable HAR," ***IEEE Consumer Electronics Magazine***.

In Preparation **Eitmam Omar Sanam** and Mohammod Abdul Motin, "Hallucination-Aware, Clinically Grounded Radiology Report Generation with Explainable AI."

Projects

November 2026 **Design and Implementation of a Camera-Based Contactless Heart-Rate Estimation System using Remote Photoplethysmography (rPPG)**

August 2026 **Design and Implementation of a Low-Cost Single-Channel EEG Headset**

June 2026 **Development and Analysis of a Custom Fine-Tuned LLM Framework for Radiology Report Generation**

December 2025 **Design and Implementation of an End-to-End Wearable Human Activity Recognition System:** novel lightweight model, custom circuit, and on-device inference

Technical Skills

Programming Language	Python, C, MATLAB
Embedded Platforms	STM32 (STM32CubeIDE, HAL), Arduino Nano 33 BLE Sense, Raspberry Pi
Frameworks	PyTorch, TensorFlow Lite for Microcontrollers, CMSIS-NN
Models	State-space models (Mamba), KANs, Transformers, LoRA/QLoRA fine-tuning of language models
Graphic Design	Adobe Illustrator, Adobe Photoshop, DaVinci Resolve, Blender
Other Software	Proteus, AutoCAD, MS Office, LOGO! Soft Comfort, emu8086, CST Studio Suite

Professional Experience

18 – 27 Feb 2026 **Industrial Trainee (Industrial Attachment),** [Certificate]
Civil Aviation Authority of Bangladesh (CAAB), Dhaka
Communication, Navigation and Surveillance (CNS) Division

- Gained hands-on exposure to the CNS infrastructure that supports safe air traffic operations.
- Studied VHF/HF radio, PABX, ILS, DVOR, PSR/SSR, ADS-B, and MLAT systems and how they work together.

Technical Training

15 – 31 Aug 2026 **Solar PV System Planning, Design, Installation and Operation,** [Certificate]
32-Hour Training Program for Engineering Students

- Conducted hands-on PV module experiments on energy conversion and electrical characteristics under varying conditions.
- Designed and implemented an end-to-end solar PV system, covering sizing, component selection, installation, and functional verification.

Extra-Curricular Activities

2026 – 2027 **Participant** in IEEE Signal Processing Cup 2027.

2025 – Present Member, IEEE RUET Student Branch

2019 – 2021 Member, ACC IT Club

References

Dr. Mohammod Abdul Motin,
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Dr. Sumaiya Kabir,
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